Understanding Stars

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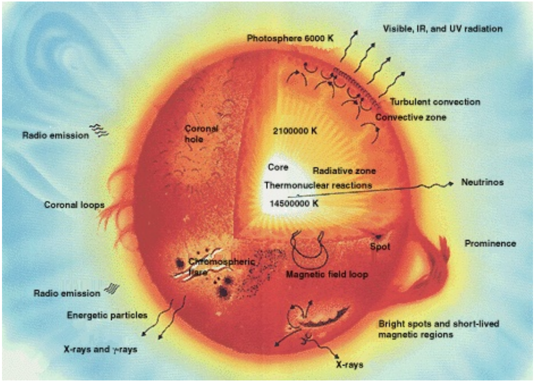
Contents

The presentation will provide answers to the following questions:

- What is a star and how does it form?
- How do stars generate energy?
- How do stars enrich the interstellar medium?
- What are the factors governing stellar structure?
- What is a Hertzsprung–Russell diagram and what can we infer from it?
- How do stars evolve over their lifetimes?
- What are the endpoints of stellar evolution?

What is a Star?

A star is a self-gravitating ball of plasma in which hydrostatic equilibrium is maintained between inward gravitational collapse and outward pressure support, sustained over timescales of by nuclear energy release in the core.



What is a Star?

Key observational properties:

- **Luminosity** L (erg s^{-1}): total power radiated.
- **Effective temperature** $T_{\rm eff}$: defined by $L = 4\pi R^2 \sigma T_{\rm eff}^4$ (Stefan–Boltzmann).
- **Mass** M: the single most important parameter governing stellar evolution.
- **Radius** R: determined by M, composition, and evolutionary state.
- **Composition**: hydrogen mass fraction X, helium Y, metals Z (where $X + Y + Z = 1$).

The Sun: $M_\odot = 1.989 \times 10^{33}$ g, $R_\odot = 6.96 \times 10^{10}$ cm, $L_\odot = 3.83 \times 10^{33}$ erg s^{-1}, $T_{\rm eff,\odot} = 5778$ K.

Star Formation

Stars form from the gravitational collapse of cold, dense regions within molecular cloud (giant interstellar structures of H_2 and dust).

A cloud fragment of mass M and temperature T collapses if it exceeds the **Jeans mass**,

$$M_J = \frac{\pi}{6}\rho\lambda_J^3, \quad \lambda_J = \left(\frac{\pi c_s^2}{G\rho}\right)^{1/2}$$

where $c_s = (k_B T/\mu m_H)^{1/2}$ is the isothermal sound speed and μ is the mean molecular weight.

Star Formation

Equivalently, collapse requires that the gravitational energy exceed the thermal energy:

$$|E_{\rm grav}| = \frac{3GM^2}{5R} > E_{\rm therm} = \frac{3Mk_BT}{2\mu m_H}$$

Checkpoint-1: Read about **Virial Theorem**.

This can be used to estimate the age of a star (like our own Sun), known as the **Kelvin–Helmholtz timescale** (**Checkpoint-2**), under certain assumptions. The age of Sun estimated using this technique comes out to be of the order of 10^7 yr, but other techniques, like Carbon Dating, has contradicting results. This implies there is something else involved as well!

Protostellar Collapse and Pre-Main-Sequence Evolution

Till now, a (sort of a) dispersed cloud of molecular Hydrogen (H_2) condensed into a more compact one, but now the atoms are very close to each other and the collisions cause the H_2-ball to heat up. Heat can't escape, so this energy goes into dissociating H_2, i.e.

$$H_2 \rightarrow 2H$$

making it a **Protostar**.

Stellar Energy Generation: Hydrogen Burning

Stars on the main sequence convert hydrogen to helium in the core via two main reaction chains:

- **Proton–proton (pp) chain** (dominant for $M \lesssim 1.3\,M_\odot$, $T_c \lesssim 1.8 \times 10^7$ K):

$$p + p \to d + e^+ + \nu_e$$
$$d + p \to {}^3\mathrm{He} + \gamma$$
$${}^3\mathrm{He} + {}^3\mathrm{He} \to {}^4\mathrm{He} + 2p$$

- **CNO cycle** (dominant for $M \gtrsim 1.3\,M_\odot$, $T_c \gtrsim 1.8 \times 10^7$ K): C, N, O act as catalysts; net result is $4p \to {}^4\mathrm{He}$ $+2e^+ + 2\nu_e + 3\gamma$.

$${}^{12}\mathrm{C}(p,\gamma) \to {}^{13}\mathrm{N}(e^+\nu) \to {}^{13}\mathrm{C}(p,\gamma) \to {}^{14}\mathrm{N}(p,\gamma) \to {}^{15}\mathrm{O}(e^+\nu)$$

$$\to {}^{15}\mathrm{N}(p,\alpha) \to {}^{12}\mathrm{C}$$

Stellar Energy Generation: Hydrogen Burning

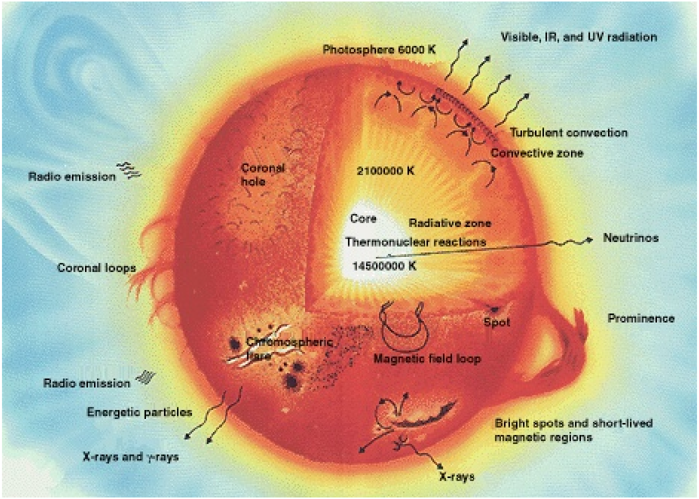


Figure: Star Structure

Stellar Energy Generation: Hydrogen Burning

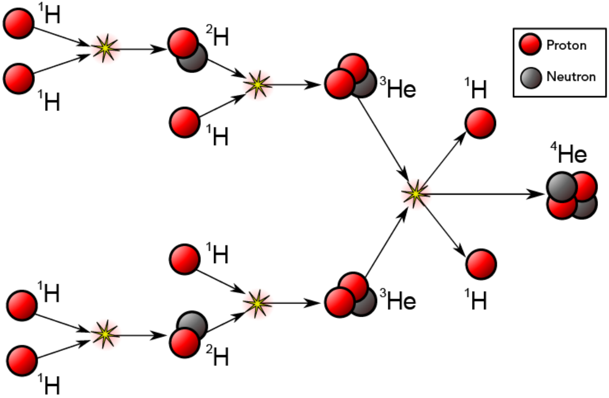


Figure: Hydrogen Burning

Stellar Energy Generation: Advanced Nuclear Burning

Once the hydrogen in the core is exhausted, more massive stars ignite successive burning stages at progressively higher temperatures:

- **Helium burning** ($T_c \sim 10^8$ K): the triple-alpha process,

$$^4\mathrm{He} + {}^4\mathrm{He} \rightleftharpoons {}^8\mathrm{Be} \xrightarrow{+^4\mathrm{He}} {}^{12}\mathrm{C}^* \rightarrow {}^{12}\mathrm{C} + \gamma$$

- **Carbon burning** ($T_c \sim 5 \times 10^8$ K):

$$^{12}C + {}^{12}C \rightarrow {}^{20}Ne + {}^4He,$$

$$\rightarrow {}^{23}Na + p,$$

$$\rightarrow {}^{24}Mg + \gamma$$

- **Neon burning** ($T_c \sim 1.5 \times 10^9$ K): photodisintegration of ^{20}Ne releases α particles that are recaptured.

Stellar Energy Generation: Advanced Nuclear Burning

- **Oxygen burning** ($T_c \sim 2 \times 10^9$ K):

$$^{16}O + ^{16}O \rightarrow ^{28}Si + ^{4}He,$$

$$\rightarrow ^{32}S + \gamma$$

- **Silicon burning** ($T_c \sim 3 \times 10^9$ K): quasi-statistical nuclear equilibrium produces an iron-peak nucleus (^{56}Ni, ^{56}Fe). Iron has the highest binding energy per nucleon making it extremely stable, so no further energy can be extracted by fusion!

Stellar Burning

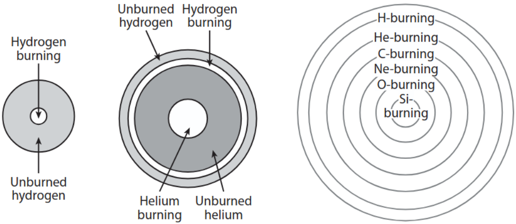


Figure: Illustration 1

Name of process	Fuel	Products	Temperature
Hydrogen-burning	H	He	60×10^6 °K
Helium-burning	He	C, O	200×10^6 °K
Carbon-burning	C	O, Ne, Na, Mg	800×10^6 °K
Neon-burning	Ne	O, Mg	1500×10^6 °K
Oxygen-burning	O	Mg to S	2000×10^6 °K
Silicon-burning	Mg to S	Elements near Fe	3000×10^6 °K

Figure: Illustration 2

Figure: Stellar Burning.

Stellar Burning

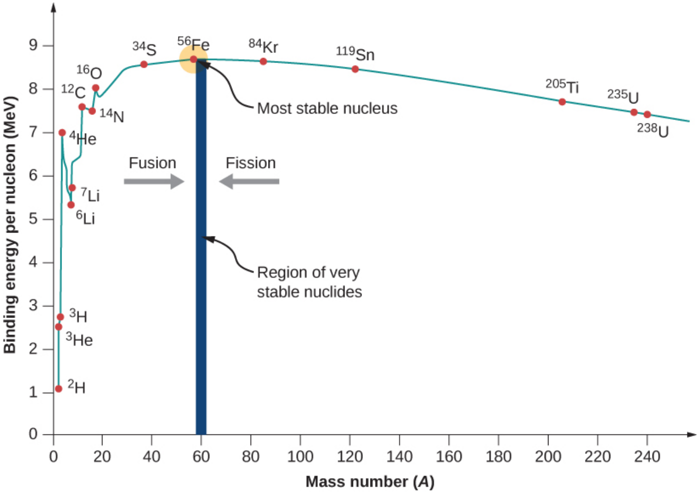


Figure: Binding energy per nucleon.

Stellar Nucleosynthesis and Chemical Enrichment

Stars are the primary factories of heavy elements in the Universe. The main production sites are:

- **Checkpoint-3** - **Big Bang nucleosynthesis**: H, D, ^{3}He, ^{4}He, Li; but nothing heavier than Be.
- **Stellar nucleosynthesis**:
  - *CNO, Ne–Ti*: hydrostatic burning in massive stars.
  - *Iron peak*: explosive nucleosynthesis in CCSN and SN Ia.
  - *s-process* (slow neutron capture): in Redgiants; produces Sr, Ba, Pb.
  - *r-process* (rapid neutron capture): in NS–NS mergers and possibly CCSNe; produces Au, Pt, Eu, Th, U.
  - *p-process*: proton capture / photodisintegration in SN shocks; produces proton-rich isotopes.
- **Cosmic-ray spallation**: Li, Be, B produced by CR interactions with CNO nuclei in the ISM.

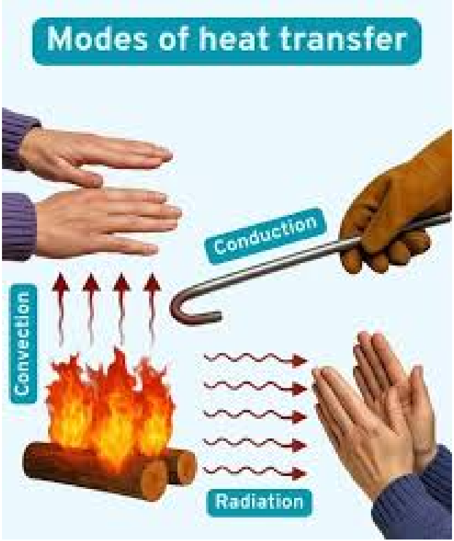


Figure: Modes of Heat Transfer.

Modes of Heat Transfer

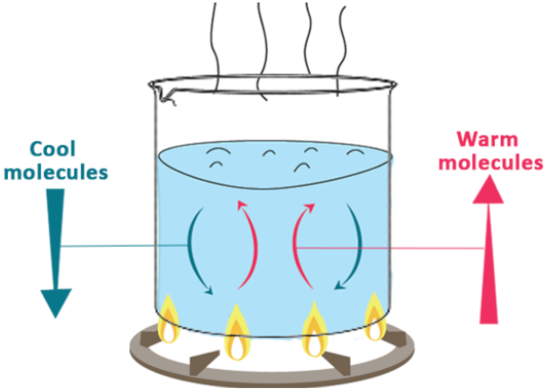


Figure: Modes of Heat Transfer.

Modes of Heat Transfer

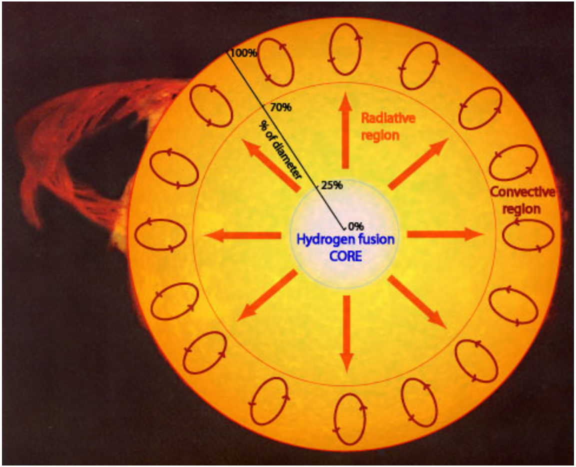


Figure: Modes of Heat Transfer in stars.

Checkpoint-4: There is convection going inside the Earth as well!

Equations of Stellar Structure

A spherically symmetric star in hydrostatic equilibrium is governed by four coupled differential equations (Eddington's stellar structure equations):

- **Mass continuity**:
$$\frac{dM_r}{dr} = 4\pi r^2 \rho$$
- **Hydrostatic equilibrium**:
$$\frac{dP}{dr} = -\frac{GM_r \rho}{r^2}$$
- **Energy generation**:
$$\frac{dL_r}{dr} = 4\pi r^2 \rho \varepsilon$$
where ε (erg g^{-1} s^{-1}) is the local energy generation rate.
- **Energy transport** (radiative):
$$\frac{dT}{dr} = -\frac{3\kappa \rho L_r}{64\pi\sigma T^3 r^2}$$; where κ is the opacity (cm^2 g^{-1})

White Dwarfs

A **white dwarf** (WD) is the degenerate remnant left after the initial burning phase. The pressure at the core a White Dwarf isn't enough to sustain a nuclear fusion reaction.

Structure:

- Supported by non-relativistic electron degeneracy pressure.
- Composition: C/O core (from He burning), thin He and H envelopes.

Chandrasekhar mass limit: The star is unstable and accretion beyond M_{Ch} in a binary triggers a **Type Ia supernova** (**Checkpoint-5**).

WDs simply cool over time, the WD luminosity function constrains the age of the Galactic disk (**Checkpoint-6**).

Type Ia Supernovae

A **Type Ia supernova** (SN Ia) is the thermonuclear explosion of a C/O white dwarf accreting mass from a binary companion or merging with another WD,

- Carbon ignites in the degenerate core at $M \to M_{\rm Ch}$; a deflagration/detonation (DDT model) incinerates the star.
- Peak luminosity: $L_{\rm peak} \approx 10^{43}$ erg s^{-1} ($M_V \approx -19.3$), powered by the decay chain ^{56}Ni $\to$ ^{56}Co $\to$ ^{56}Fe.

The 1998 discovery of cosmic acceleration via SN Ia was awarded the 2011 Nobel Prize in Physics.

Neutron Stars and Pulsars

The remnant of a Core-Collapse Supernova (CCSN) (for progenitors $8 \lesssim M/M_\odot \lesssim 20$) is a neutron star (NS), supported by neutron degeneracy pressure and nuclear forces. The density of a Neutron Star is like compressing an entire mountain to the size of a teaspoon!

Pulsars: rapidly rotating NS with strong magnetic fields ($B \sim 10^{12}$–10^{13} G). The rotating dipole emits beamed electromagnetic radiation; if the beam sweeps past Earth, we observe regular pulses.

Pulsars

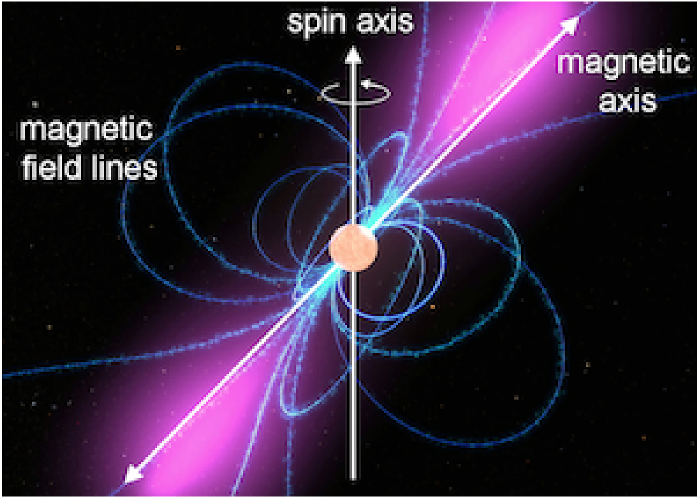


Figure: Pulsars

Black Holes

For very massive progenitors ($M \gtrsim 20$–$25\, M_\odot$, depending on metallicity and mass loss), the remnant mass exceeds the maximum NS mass and a black hole forms.

A non-rotating BH is characterised solely by its mass M (Schwarzschild metric):

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

with Schwarzschild radius $r_s = 2GM/c^2 \approx 3\,(M/M_\odot)$ km.

Figure: M87

Stellar Atmospheres and Spectral Classification

The stellar spectrum encodes the photospheric temperature, composition, pressure (gravity), etc. The Harvard classification organises stars by decreasing $T_{\rm eff}$:

Class	$T_{\rm eff}$ **(K)**	**Colour**	**Characteristic lines**
O	$>$ 30,000	blue-white	He II, N III, O III ionised
B	10,000–30,000	blue-white	He I, H Balmer
A	7,500–10,000	white	H Balmer strongest
F	6,000–7,500	yellow-white	Ca II H&K, metals
G	5,200–6,000	yellow	Ca II H&K, Fe, CN
K	3,700–5,200	orange	Ca II, TiO begins
M	$<$ 3,700	red	TiO, H_2O bands

The mnemonic: *Oh Be A Fine Girl/Guy, Kiss Me!*

Stellar Atmospheres and Spectral Classification

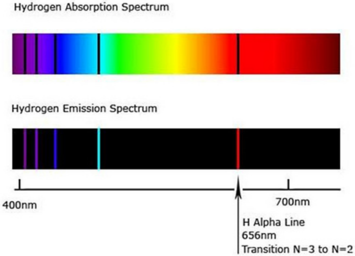


Figure: Illustration 1

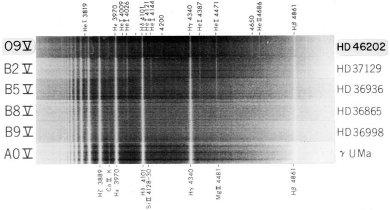


Figure: Illustration 2

Figure: Spectrum.

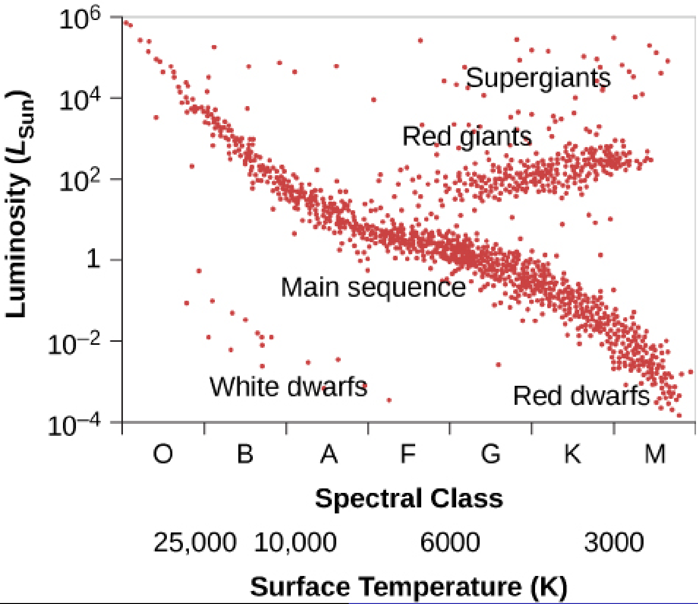
The Hertzsprung–Russell Diagram

The Hertzsprung–Russell (H–R) diagram plots stellar luminosity L vs. effective temperature T_{eff} (or equivalently, absolute magnitude vs. colour index). It is the central organising diagram of stellar astrophysics.

Major features:

- **Main sequence (MS)**: a diagonal band from hot, luminous O stars to cool, dim M dwarfs. Stars on the MS are burning H in their cores.
- **Giant branch**: luminous, cool stars ($T_{\mathrm{eff}} \sim 4000$–$5000$ K) with expanded envelopes.
- **Horizontal branch (HB)**: core He-burning at roughly constant $L \sim 50\,L_\odot$.
- **White dwarf sequence**: faint, hot, compact remnants cooling at constant radius.
- **Instability strip**: vertical band containing pulsating variables (Cepheids, RR Lyrae).

The Hertzsprung–Russell Diagram



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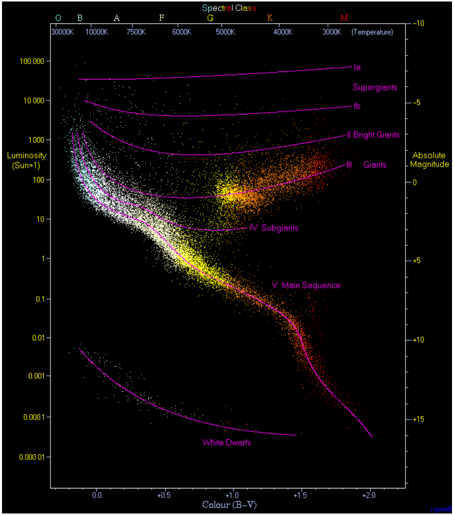

Figure: HR diagram.

The Hertzsprung–Russell Diagram

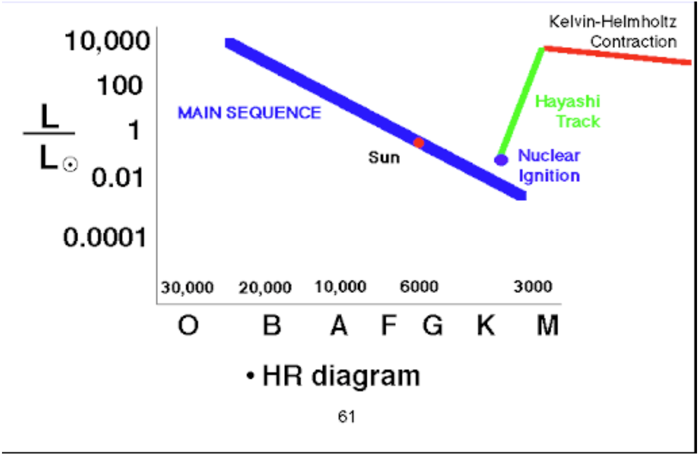

Figure: HR diagram.

Following H exhaustion in the core, the evolutionary sequence is:

- **Subgiant branch**: H-exhausted core contracts and heats; H-burning shell ignites; envelope expands and cools.
- **Red Giant Branch (RGB)**: star ascends the giant branch at roughly constant $T_{\rm eff}$; deepening convective envelope (first dredge-up) brings CN-cycle products to the surface.
- **Helium flash**: for $M \lesssim 2.3\,M_\odot$, the degenerate He core ignites in a violent, off-centre thermal runaway at $L_{\rm He} \sim 10^{11}\,L_\odot$ (over seconds). Electron degeneracy is lifted; star settles onto the horizontal branch.

Low- and Intermediate-Mass Stellar Evolution ($M \lesssim 8\,M_\odot$)

Following H exhaustion in the core, the evolutionary sequence is:

- **Horizontal Branch (HB)**: core He burning + shell H burning. Duration $\sim 10^8$ yr.
- **Asymptotic Giant Branch (AGB)**: He and H burning shells alternate (thermal pulses). Third dredge-up brings ^{12}C to the surface; star may become carbon-rich (C star).
- **Mass loss**: strong AGB winds ($\dot{M} \sim 10^{-5}$–$10^{-4}\,M_\odot\ \mathrm{yr}^{-1}$) eject the envelope, producing a planetary nebula and leaving behind a white dwarf.

High-Mass Stellar Evolution ($M \gtrsim 8\,M_\odot$)

Massive stars evolve rapidly and end their lives explosively:

- **Main sequence**: CNO-cycle burning in a large convective core; strong radiation-driven winds ($\dot{M} \sim 10^{-6}$–$10^{-5}\,M_\odot$ yr^{-1}) strip mass progressively.
- **Post-MS**: star expands to a red supergiant (RSG, $T_{\rm eff} \sim 3500$ K, $R \sim 10^3\,R_\odot$) or, for the most massive, an LBV (Luminous Blue Variable) or Wolf–Rayet (WR) star (stripped envelope, bare He or C/O core).
- **Onion-shell structure**: successive burning stages (He, C, Ne, O, Si) occur in nested shells on rapidly decreasing timescales. Silicon burning lasts only ~ 1 day.

High-Mass Stellar Evolution ($M \gtrsim 8\,M_\odot$)

Massive stars evolve rapidly and end their lives explosively:

- **Iron core**: once an $\sim 1.4\,M_\odot$ ($M_{\rm Ch}$) iron core builds up, electron degeneracy pressure can no longer support it. Gravitational collapse begins.
- **Core-collapse supernova** (CCSN): the inner core reaches nuclear density ($\rho \sim 3 \times 10^{14}$ g cm^{-3}) and bounces. A shock wave propagates outward, revived by $\sim 10^{53}$ erg of neutrino energy deposition, and unbinds the stellar envelope.

Summary

- Stars are self-gravitating plasma spheres sustained by nuclear fusion, governed by four coupled differential equations of stellar structure: mass continuity, hydrostatic equilibrium, energy generation, and energy transport.
- The mass of a star is the dominant parameter. It determines luminosity ($L \propto M^4$), lifetime ($t_{\mathrm{MS}} \propto M^{-2.5}$), and final fate (WD, NS, or BH).
- The H–R diagram organises all evolutionary stages; spectral classification (OBAFGKM) links observed colours and lines to T_{eff} and $\log g$.

Summary

- Low-mass stars die as planetary nebulae + white dwarfs; massive stars undergo core-collapse supernovae, leaving neutron stars or black holes.
- Stars are the engines of cosmic chemical evolution: from primordial H and He, they have built up the periodic table through hydrostatic and explosive nucleosynthesis, s-process, and r-process.

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Thank You!

Questions?

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Figure: QR 1

Figure: QR 2

Figure: Join *Astrae*.